

Assessing nature-based solutions

Nature-based solutions (NBS) are interventions designed to use the power of nature to address multiple sustainability challenges and contribute to biodiversity. They are increasingly recognized as a crucial means through which the intertwined challenges of climate change, the loss of biodiversity and social injustice can be addressed. For instance, at the global level NBS have been recognized as important tools to achieve the Sustainable Development Goals, to address climate change under the UN Framework Convention on Climate Change, and to meet the goals of the Kungming-Montreal Global Biodiversity Framework of the Convention on Biological Diversity. Within the European Union, NBS play a role in the EU Biodiversity and Adaptation strategies.

There have been various planning tools that can be used to design and implement NBS projects. Perhaps most notable among these is the [IUCN Global Standard for Nature-based Solution](#), which provides a robust checklist of 8 criteria with 28 indicators with guiding questions that can be used to plan an NBS project. Many of these indicators are process-level attributes that can only be assessed by those with local knowledge, often those who are planning and implementing the NBS project. For example, criteria 1.3 is that “Human well-being outcomes arising from the NBS are identified, benchmarked and periodically assessed”.

However, as policymakers and the private sector increasingly fund NBS (and many older parks and conservation programs are adopting the term NBS), there is a need for a way to rapidly assess sets of hundreds or thousands of NBS, in a variety of diverse political, social, and ecological contexts. What constitutes a high-quality NBS, given the variety of goals and forms that an NBS can take, and how can the set of NBS in use in a city or country be consistently assessed?

The Naturescapes scorecard

As part of the [Naturescapes](#) collaboration, we have developed a scoring methodology that is meant to close this knowledge gap. The purpose of this document is to briefly describe the draft methodology, so that others can learn from our experiences. Note that this methodology is under review and may change.

The overall Naturescapes project studies NBS from a landscape perspective, asking the question as to whether and how multiple NBS at a landscape scale can create synergistic outcomes that address long-standing socio-economic challenges, enhance biodiversity and respond to climate change. We study diverse social, cultural, economic and geographical settings across 30 functional urban areas (FUAs) with a

specific focus on 12 case studies in the European Union, Latin America and the United States of America. As part of that work, Naturescapes researchers have mapped and characterized 390 NBS across these 30 FUAs.

The purpose of the Naturescapes scorecard is to provide an overview of the contributions of the 390 NBS sites to address sustainability challenges based on a common set of quantitative and qualitative indicators. Of these 390 NBS sites, 38 were city-wide interventions (e.g., city-wide open space plans), and are not scored at all. Another 24 did not have precise spatial boundaries or were otherwise not suitable for GIS analysis and so were not scored for the GIS-derived variables but were scored for the qualitative indicators.

The scorecard aims to serve as a simple starting point for understanding individual NBS impacts and the impacts across NBS assemblages (which we call “naturescapes”) in specific thematic domains or areas. Our scoring methodology is designed to be applicable to almost any NBS with globally available data and with a review of electronically accessed commonly available public documents about an NBS project.

The audience for the scorecard of these 390 NBS includes local and regional policymakers and practitioners (e.g., planners and landscape architects), as well as those in the private sector investing in NBS. Users of the scorecard data will also include researchers in the Naturescapes project. More broadly, the methodology we develop here will be useful to the broader academic and policy community interested in NBS evaluation. We believe this scoring methodology can help actors to understand better the variety of project impacts that NBS can have in relation to climate, biodiversity and communities across urban, rural and coastal environments.

Beyond its analytical purpose for the Naturescapes project, the methodology used for the scorecard has the potential to be applied in assessing other NBS not considered within the Naturescapes project— helping cities, investors, and policymakers identify where interventions can deliver the highest ecological and social returns. By linking biophysical, socio-economic, and governance indicators, the scorecard provides a structured way to assess NBS performance, using terminology that should be consistent with emerging market and regulatory frameworks such as the European Central Bank’s climate factor for collateral risk assessment and the EU’s forthcoming Nature Credits standards.

Framework

Themes

Our scoring methodology considers four interrelated thematic areas: Climate, Biodiversity, Social Justice, and Transformative Potential. Under each theme, we identified a set of indicators to illustrate spatially explicit environmental outcomes and broader socio-economic and strategic results of NBS implementation.

- Under the **Climate** theme, indicators assess the contribution of NBS assemblages to climate change mitigation and adaptation through impacts on land surface temperature reduction, carbon storage, and stormwater retention and evaluate these contributions compared to stated climate-related ambitions.
- The **Biodiversity** theme evaluates potential NBS impacts on biodiversity in terms of habitat protection and ecosystem connectivity, as well as comparing performance against declared biodiversity goals.
- The **Social Justice** theme focuses on the accessibility and equitable distribution of NBS benefits, including an evaluation of the proximity of people to green and blue spaces, recreational opportunities, and the inclusiveness of beneficiary groups. It also evaluates these impacts from the perspective of social justice-related ambitions.

Lastly, the **Transformative Potential** theme considers the long-term sustainability and systemic impact of NBS by assessing the quality of project design, integration of long-term and cross-cutting perspectives, and considerations given to stakeholder diversity.

Conceptual approach

Our scoring methodology recognizes that since NBS are multifunctional, multiple attributes must be incorporated in a score to have an overall sense of their performance. As the measurement of attributes may vary depending on their character and data limitations, bringing the individual measures to the same comparable level requires normalization according to a scoring framework.

From the procedural perspective, scoring involves the following: (a) development of a scoring framework, with performance categories and thresholds between the categories; (b) identification of relevant impact attributes to be scored; (c) the selection of indicators related to these attributes; (d) measuring the indicators based on quantitative or qualitative data; (e) relating measured indicator values to a relevant reference value; (f) and the determination of the score for the indicator, based on where actual performance falls in comparison with the reference value and what score category it corresponds to in the scoring system.

Scoring may or may not involve aggregation of component scores into sub-aggregates or a single score, which requires the consideration of the aggregation approach (e.g., additive score or average of individual score components) and weighting (equal or non-equal). The selection of both scoring and aggregation approaches involves trade-offs: decision-makers often prefer single aggregate scores for simplicity, but a single aggregate index may conceal key details. For our Naturescapes scorecard, we chose to present the scores for all the indicators, as well as sub-aggregates for the four themes, but to avoid presenting an overall aggregate score. As shown below, the themes reflect quite different facets of NBS performance, and it seemed unhelpful to hide these different facets behind a single score of NBS performance.

Indicator selection

The Naturescapes scorecard aims to communicate the potential contributions of NBS interventions across our four key thematic areas, using a combination of quantitative and qualitative indicators that describe the potential impact of NBS on a particular aspect of these four thematic areas.

The indicators were applied to the assessment of the 390 NBS collected in the Naturescapes NBS database, now incorporated into the broader [Urban Nature Atlas](#). Indicators based upon geospatial data were applied to the NBS based on locations, while qualitative indicators were calculated based on data available collected for the NBS database, primarily from electronic documents available on the Internet.

During indicator selection, an interdisciplinary team from different Naturescapes project partners (CEU, TNC, LU, WWF, DU) worked collectively to define a list of 16 indicators that can characterize the complex and multifaceted impact of NBS. Given the great diversity of 390 NBS across 30 cities and the multitude of diverse impacts and measurement options, the indicators in the scorecard do not represent a complete set and should be considered illustrative. Future users of our methodology are encouraged to add other indicators based upon local priorities and context.

Indicator selection was guided by a set of selection criteria that align with the objectives of the scorecard:

- **Relevance and thematic suitability:** Alignment with the four thematic areas of the scorecard evaluation framework.
- **Data availability:** Practical considerations around data accessibility played a critical role in the selection process. Indicators were favored when reliable data was already available or could be obtained for almost all NBS. Where data was not available for a particular indicator for a particular NBS, it is scored as “Not Available” on the scorecard.

- **Alignment with project elements:** Consideration of parallel or complementary activities in other Naturescapes work packages to leverage synergies and avoid duplication. The overall [Naturescapes](#) project explores how different approaches to working with nature, such as restoring wetlands, planting trees, or managing land sustainably, can come together to benefit both people and the planet.
- By bringing together researchers, NGOs, and consultants from different fields, Naturescapes aims to uncover how combining nature-based approaches and solutions can lead to real, lasting change across landscapes.
- **Scale:** Suitability of indicator for application at the individual NBS level, while also allowing for potential aggregation and comparative analysis at the nature- or FUA-level.
- **Data output:** Suitability of the indicator to produce consistent quantitative or semi-quantitative data for a large sample of NBS.

Based on these indicator selection criteria, four indicators were selected for each key thematic area, as presented in Table 1.

Table 1: List of selected scorecard indicators

Indicator	Indicator description
Climate	
Land surface temperature	Summer land surface temperature (Celsius) average within the NBS sites relative to a 1 km buffer around sites.
Carbon storage	Carbon density average within the NBS site relative to a 1 km buffer
Stormwater holding capacity	Estimation of how much more rain (mm rain) an area can infiltrate with the NBS than without it
Climate-related ambition and performance	Evaluates whether the NBS project aimed to address problems related to climate action (for adaptation, resilience, and/or mitigation) and delivered project impacts in this area.
Biodiversity	
Protected Areas	Degree of protection in the NBS, based upon the presence of a protected area and the IUCN protected category .
Coastal habitats	Coastal habitat fraction within the low-elevation coastal zone
Green/blue area fraction	% increase of land covered by green/blue habitats by the NBS vs a 1 km buffer
NBS biodiversity-related ambition and performance	Evaluates whether the NBS project aimed to address biodiversity-related problems (restoration and/or conservation) and delivered project impacts in this area.
Social Justice	
Population access to blue and green space	Population within 1 km of NBS sites
Recreation potential	Percent increased accessibility compared to having no NBS in a landscape with a 1 km radius from the border of the NBS.
Inclusiveness of project beneficiaries	Evaluates the extent to which an NBS project identifies and supports a diversity of beneficiaries, with greater recognition for NBS that include citizens, community groups, or explicitly address marginalized or vulnerable groups.
NBS social justice-related ambition and performance	Evaluates whether the NBS project aimed to address problems related to social justice & cohesion and delivered project impacts in this area.

Transformative Potential	
Potential for high-quality project result delivery	The high-quality potential of an NBS project is calculated based on a combination of different NBS characteristics, including the presence of participatory methods or forms of stakeholder engagement, multiple impacts delivery (climate, biodiversity, and social justice), delivery of equitable impacts, and demonstrated NBS project transparency.
Long-term perspective	Evaluate whether an NBS project is framed as an intervention that considers future or long-term pressures, such as being capable of withstanding future environmental stressors (e.g. climate hazards), or ensuring long-term effectiveness (e.g. maintenance costs).
Diversity of stakeholder involvement	Reflection on the stakeholder groups involved across different stages of the NBS project cycle.
Alignment of NBS Targets with Climate, Biodiversity, and Social Objectives	Evaluates whether an NBS targets problems across the key areas of climate action, biodiversity, and social justice.

Scoring approach

This section provides a detailed overview of the approaches taken for indicator scoring for both quantitative and qualitative indicator types, as well as how scores are aggregated. Many more technical details are available in the indicator notes document that will be publicly available after the review and revision process is completed.

Quantitative variables

In general, we measure indicator values for quantitative indicators as the relative increase of service provision that can be attributed to an NBS compared to a reference condition. Often, when there is no clear counterfactual, we compare NBS averages with the average within a 1km buffer, the surroundings of an NBS. For a few indicators where there was a clear and tractable counterfactual scenario, the reference condition is the counterfactual without the NBS presence. To express indicators in a common (normalized) unit that can be compared against one another, we classify the raw indicator values into a four-point scale. Score 1 includes exceptional cases of no service provision at all (i.e., zero values on the raw indicators). For all other sites with non-zero values of the raw indicator (those with a measured effect), we classify them into three equal sized groups based on tertiles across all our observed NBS sites (scores 2-4).

Qualitative variables

Our approach is based on a structured methodology designed to translate NBS-related narrative and categorical data reported in the Naturescapes NBS database into

relevant NBS indicators, standardized according to the same 4-level scoring scale as that applied to quantitative geospatial indicators. This method enables comparability across all NBS projects included in the database while remaining sensitive to differences in project scope, reporting depth, and data availability.

Each indicator is constructed to assess both the extent (e.g., the number of NBS project beneficiaries) and the intention or depth (e.g., who benefits, inclusion of supporting amenities in NBS planning) of a specific project characteristic, such as long-term planning or project inclusiveness.

The 4-point scale is applied to qualitative indicators as follows, using the indicator “Alignment of Climate-related Ambition and Performance” as an illustrative example:

- Score 1: No climate-related ambition and performance - Neither are climate goals set, nor does the project report climate impacts.
- Score 2: Climate-related ambition without performance - Climate goals are established, but there is no evidence of climate impacts being reported.
- Score 3: Climate-related performance without a clear initial ambition - No climate goals have been established, but climate impacts are being reported.
- Score 4: Aligned climate-related ambition and performance - Climate goals have been established, and climate impacts have been reported.

For instance, the indicator “Inclusiveness of project beneficiaries” evaluates whether a project identifies and considers the needs of those who will directly benefit from the intervention. Projects that exclusively report institutional beneficiaries (e.g. government agencies or public sector institutions) receive a lower score (score 2), while those that include a mix of intended beneficiaries with a particular focus on community-oriented beneficiaries (e.g. citizens, children, food producers) score higher (score 3). Projects that, in addition to including community-oriented beneficiaries, explicitly engage marginalized or vulnerable groups (e.g. low-income communities, indigenous peoples, or coastal-dependent populations) receive the highest score (score 4), reflecting a deliberate effort to address social equity. In the case of this specific indicator, the lowest score (score 1) would be attributed to an NBS project that does not report any intended project beneficiary.

In the qualitative assessment of NBS projects, a mix of single-variable and composite indicators is applied. Some indicators, such as the “Inclusiveness of project beneficiaries,” are based on a single, clearly defined data point included in the Naturescapes NBS database. Others, such as the “Long term perspective” indicator, are composite, combining two or more related variables (e.g., clear expression of “long-

term perspective of the NBS”, and the “presence of an assessment, evaluation and/or monitoring process”) to provide a more comprehensive assessment.

Aggregating scores for themes

We aggregated multiple indicators across a given thematic area into one composite score. Individual indicator scores, expressed on a standardized 1 to 4 scale, are aggregated to reflect the overall score for the theme.

The aggregation approach taken in the scorecard is equally weighted, meaning that each indicator contributes equally to the final score. This approach captures a project's cumulative effort across multiple indicators, without favoring particular indicators.

Table 2: Example of scoring approach for the social justice theme for one NBS site

Indicator	Score
Population access to blue and green space	2
Recreation potential within and without NBS in a 1 km buffer	3
Inclusiveness of project beneficiaries	1
NBS social justice-related ambition and performance	2
Aggregated score (4-16)	8

Expected output

Currently, our scorecard methodology and the calculated scores for our 390 NBS sites are in review. Once that review is finished, the technical indicator notes, and the scorecard results will be available on the Commission website. In 2026, the scorecard results will be presented on a publicly accessible, interactive website, along with a companion policy brief presenting and analyzing trends in our scorecard dataset. Website functionality and design will be finalized after consultation with those in the target user group. Currently, the plan is to create a map interface that allows users to visualize the location of all NBS on top of a suitable basemap, which can be changed. By clicking on an NBS, users can view its scores. Users will be able to select multiple NBS, either by clicking on multiple sites or by selecting multiple NBS with a common indicator value (e.g. all NBS that have climate adaptation as a goal) and then can see summary statistics as well as a comparative table for that set of NBS.

How others can use our approach

We share our draft methodology of the Naturescapes scorecard so that others can learn from our approach. As the methods and the scores for the 390 NBS sites are still under review, they may change slightly. However, what is important about our methodology for others considering ways to assess NBS is likely our overall approach, rather than the technical details of the specific indicators we picked. We found that it is possible to pick four major themes of policy importance when evaluating NBS (biodiversity, climate, social justice, and transformative change) and then develop indicators that are objective and can be assessed for data generally available. For GIS-derived variables, that means using globally available datasets, which are of sufficient quality to support some meaningful indicators. For contextual variables, that means designing indicators that can be assessed based upon public information on websites about the goals and structure of NBS projects.

Recently, there has been much attention by policymakers and corporate/financial decision makers to assessing the quality of NBS projects, across multiple dimensions, and we hope our scoring methodology shows that simple, multi-variable scoring methods can be used to assess large samples of hundreds or thousands of NBS.